LITERATURE REVIEW

From Existing Research to Our Idea

Intelligent Energy Management (2024)

Contribution

Random Forest for appliance energy prediction.

Limitation

Focused on the AI model only — no complete heterogeneous IoT platform with multiple communication technologies.

Smart Home Energy Optimization System (2024)

Contribution

LSTM-based energy optimisation with a three-layer architecture.

Limitation

Described the framework only — did not implement a real smart home with Zigbee, LoRa, BLE and an edge gateway.

Prediction and Optimization (2023)

Contribution

Compared forecasting models (VAR, Prophet, LightGBM) for smart-home energy prediction using IoT data.

Limitation

Focused on algorithms rather than a complete system.

Research Gap: None of these studies integrated heterogeneous IoT communication, AI prediction, and security into one working platform.:

TECHNOLOGY CHOICES

Design Decisions

Zigbee

Low-power mesh network for reliable indoor sensor communication.

LoRa

Long-range communication for distant nodes such as the garage.

BLE

Secure short-range communication between the smartphone and the smart lock.

Raspberry Pi

Edge gateway for protocol integration and local processing.

MQTT

Lightweight publish–subscribe protocol for reliable real-time IoT communication.

Bi-LSTM

Time-series forecasting of energy use to drive optimisation.

Physical and Network Architecture



Room 1
- Zigbee

















Room 2
- Zigbee





Edge Gateway (Raspberry Pi 5)
LoraWAN
- Zigbee



Room 3 (Garage)
- LoRAWAN
- BLE

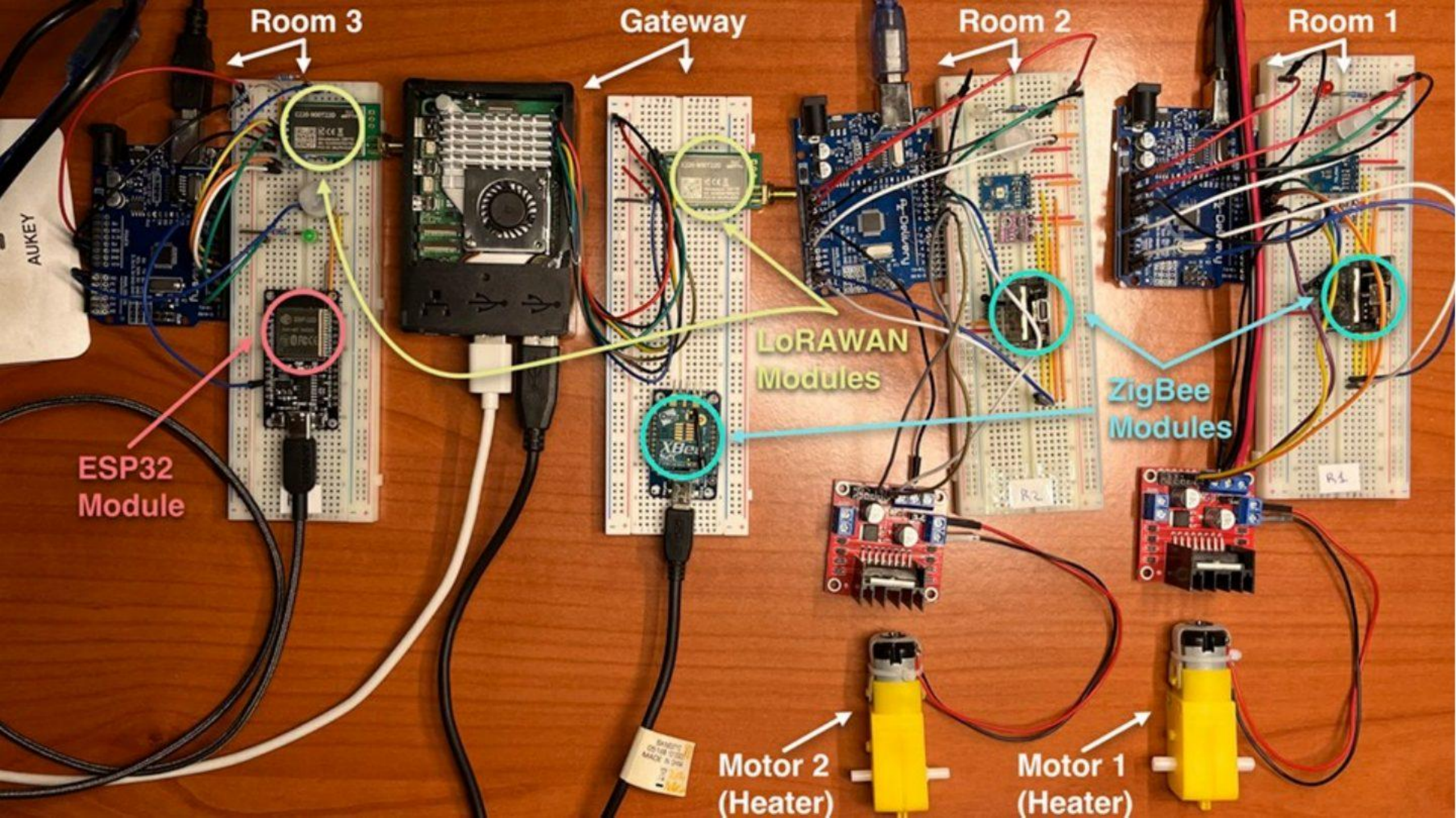
AI Energy Prediction (Bi-LSTM)

MQTT








Room 3
Gateway
Room 2
Room 1
LoRAWAN
Modules
ZigBee
Modules
ESP32
Module
Motor 2
(Heater)
Motor 1
(Heater)

My Technical Contributions

1

Integrated platform

Integrated the Flutter application, BLE communication and MQTT with the Raspberry Pi edge gateway.

2

AI energy prediction

Collaborated in integrating the AI prediction module into the overall system.

3

Flutter app

Developed the Flutter mobile application for real-time monitoring and smart home control.

4

Security extension

Designed and implemented the secure BLE communication between the Flutter application and the smart lock.

PROJECT EVOLUTION

Why security became part of the project

1

PHASE 1

Integrated Smart Home Platform

Built the heterogeneous IoT platform with AI-based energy prediction — the original goal.

2

THE TRIGGER

IoT Security course

Learning attack surfaces made one thing clear: the smart lock is the most safety-critical device — and it was exposed.

3

PHASE 2

End-to-End Secure Smart Lock

Redesigned the smart lock with authentication + end-to-end encryption, then validated it against real attacks.

CHALLENGES

Engineering Challenges I had to solve



Challenge: Integrating heterogeneous communication protocols.
Solution: Unified Zigbee, LoRa, BLE and MQTT through a Raspberry Pi gateway.



Challenge: Reliable real-time communication.
Solution: Used MQTT publish/subscribe messaging.



Challenge: Secure smart lock communication.
Solution: Implemented secure BLE authentication.



Challenge: LoRa frequency mismatch during debugging.
Solution: Validated the hardware configuration and communication experimentally.



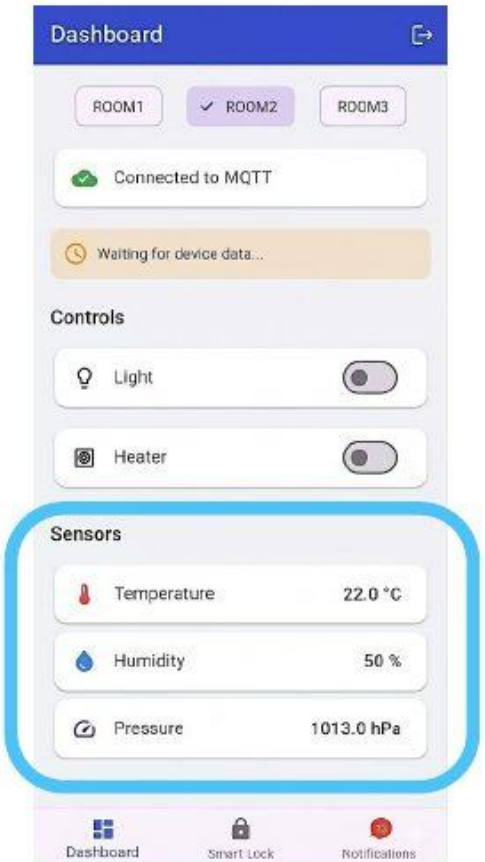
Challenge: AI integration with the platform.
Solution: Connected the AI prediction module to the edge gateway for real-time inference.

RESULTS

System Demonstration



```
home/status {"room": "room2", "temperature": 24.1, "humidity": 55.8, "pressure": 1007.8}
home/status {"room": "room1", "temperature": 24.7, "humidity": 54.2, "pressure": 1006.8}
home/status {"room": "room2", "temperature": 24.1, "humidity": 55.7, "pressure": 1007.8}
home/status {"room": "room1", "temperature": 24.7, "humidity": 54.1, "pressure": 1006.9}
```

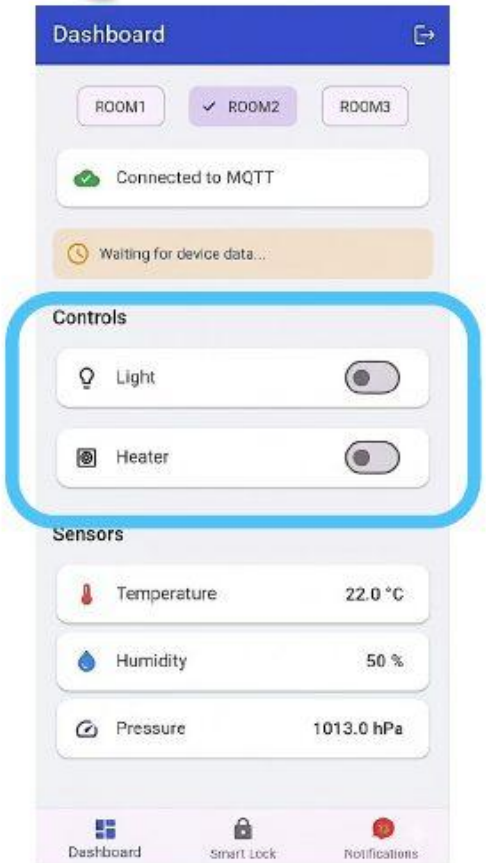


Real-Time Data Synchronization via MQTT





```
home/room1/command {"heater": "on"}
home/room2/command {"heater": "on"}
home/room1/command {"light": "off"}
```



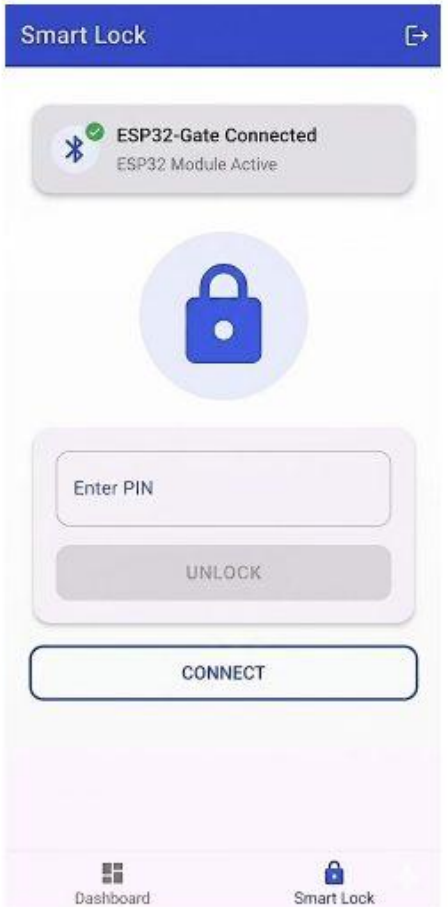
Remote Device Control

BLE Smart Lock Interface

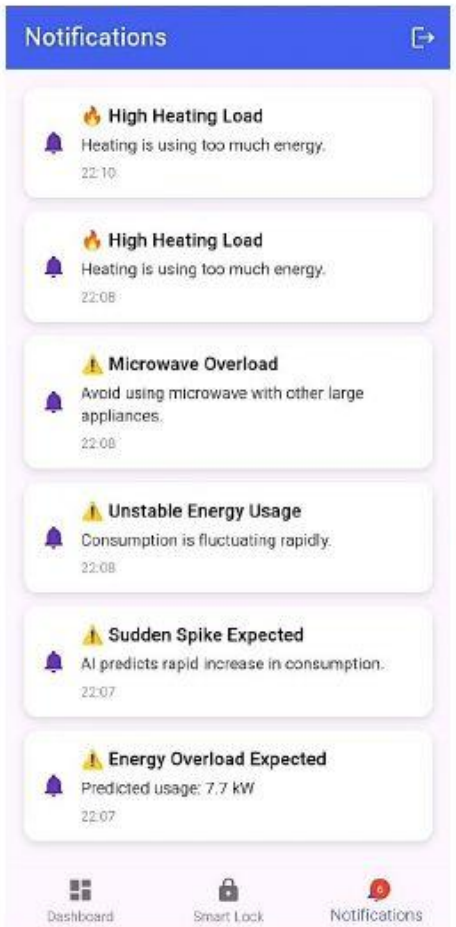
If the entered PIN matches the stored authentication code, the ESP32 activates the output signal connected to the Arduino node, which subsequently triggers the garage unlock event and sends a confirmation message to the Gateway via the LoRa network.







AI-Based Alerts and Notifications




```
WARNING: Load > 2.5 kW. Trip in 0 seconds.
>> CRITICAL ALERT: Main Breaker Warning | Sustained overload: 3.68 kW. Initiating emergency cutoff.
AUTO ACTION: room1 | {'heater': 'off'}
[ZIGBEE SENT] r1 heater off
```

